

Further Integration

Key topics

Integration by parts

Integration by substitution

Recap

Integration is the reverse of differentiation.

Derivative $\frac{dy}{dx}$ = Slope of function

Integral $\int_{start}^{end} dx$ = Area under the curve

x	$\frac{1}{2}x^2$
x^n	$\frac{1}{n+1}x^{n+1}$
$\frac{1}{x}$	$\ln(x)$
e^x	e^x
e^{kx}	$\frac{1}{k}e^{kx}$
$\sin(x)$	$-\cos(x)$
$\cos(x)$	$\sin(x)$
$\sin(kx)$	$\frac{-\cos kx}{k}$
$\cos(cx)$	$\frac{\sin kx}{k}$
$\tan x$	$\ln(\sec x)$

Reminder

Evaluate: $\int_0^2 3x^3 \, dx$

$$\int_0^2 3x^3 \, dx = \left[\frac{3}{4} x^4 \right]_0^2$$

$$= \frac{3}{4} (2^4 - 0^4) = 12$$

Your turn, evaluate: $\int_1^2 2x^{\frac{1}{2}} - 18x \, dx$

$$\int_1^2 2x^{\frac{1}{2}} - 18x \, dx = \left[\frac{4}{3} x^{\frac{3}{2}} - 9x^2 \right]_1^2$$

$$\text{Top: } \left(\frac{4}{3} 2^{\frac{3}{2}} - 9 \times 2^2 \right) = -32.23$$

$$\text{Bottom: } \left(\frac{4}{3} 1^{\frac{3}{2}} - 9 \times 1^2 \right) = -7.67$$

$$\text{Total: } -32.23 + 7.67 = -24.56$$

Products of functions

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

We looked at adding and subtracting functions, what about functions which are multiplied?

$$\int (x^2 + 7) \sin x \, dx$$

$$\int \frac{2x}{e^x} \, dx$$

$$\int f(x) g(x) \, dx$$

Integration by parts

We have to make use of a formula to be able to integrate combined functions:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

It's convoluted (and I'm sorry that it uses u and v again!)

We let one function equal u and one equal $\frac{dv}{dx}$

Derivation

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}, \text{ where } y = u(x)v(x)$$

$$u \frac{dv}{dx} = \frac{dy}{dx} - v \frac{du}{dx}$$

$$\int u \frac{dv}{dx} = \int \frac{dy}{dx} - \int v \frac{du}{dx}$$

$$\int \frac{dy}{dx} = y = uv$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

Simple example: $\int x \sin x \, dx$

$$\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$$

We let $u = x$ and $\frac{dv}{dx} = \sin x$

We integrate $\frac{dv}{dx}$, to get v and differentiate u , to get $\frac{du}{dx}$:

$$\int \frac{dv}{dx} \, dx = \int \sin x \, dx = -\cos x + c$$

$$\frac{du}{dx} = \frac{d}{dx} x = 1$$

Workings

$$u = x \quad \frac{dv}{dx} = \sin x \quad \frac{du}{dx} = 1 \quad v = -\cos x + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int u \frac{dv}{dx} dx = (x)(-\cos x + c) - \int (-\cos x + c)(1) dx$$

$$= -x \cos x + cx - (-\sin x + cx + d)$$

$$= \sin x - x \cos x - d$$

Harder example

Consider $\int x^2 \cos x \, dx$

$$u = x^2 \qquad \frac{dv}{dx} = \cos x \qquad \frac{du}{dx} = 2x \qquad v = \sin x$$

$$\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$$
$$\int u \frac{dv}{dx} \, dx = x^2(\sin x) - \int (\sin x)(2x) \, dx$$

The term on the right STILL has a function multiplied by a function!

In this case $\int (\sin x)(2x) \, dx = 2 \int x \sin x \, dx$,

this is the simple problem we just did, giving:

$$x^2(\sin x) - 2(\sin x - x \cos x) + c$$

A special case

Consider $\int \ln x \, dx$. This isn't on our list of standard results. We can treat it as a product of two functions,

$$1 \times \ln x$$

$$u = \ln x \qquad \frac{dv}{dx} = 1 \qquad \frac{du}{dx} = \frac{1}{x} \qquad v = x$$

$$\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$$

$$\int u \frac{dv}{dx} \, dx = x \ln x - \int x \frac{1}{x} \, dx$$

$$x \ln x - x + c$$

Interpretation

- 1) We can actually ignore the constant of integration and just add a constant at the end of the process
- 2) We may need to apply the formula more than once to successfully integrate a product of functions
- 3) Our choice of which function is u and which is v matters – we have to be able to differentiate u easily and integrate v easily

Function selection

Although we are selecting u and $\frac{dv}{dx}$, we want to keep our eye on v and $\frac{du}{dx}$

If $v \frac{du}{dx}$ is no easier to integrate than $u \frac{dv}{dx}$, we are wasting our time

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

Single powers of x are great targets for u , because they become constants when we differentiate

Definite integrals

We can use limits, just like we did before and they work very much as you'd expect:

$$\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b v \frac{du}{dx} dx$$

This means that we can evaluate the product of two functions as we would any other integral

Definite integral example

Find: $\int_0^2 x e^{-x} dx$

$$u = x \qquad \frac{dv}{dx} = e^{-x} \qquad \frac{du}{dx} = 1 \qquad v = -e^{-x}$$

$$\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b v \frac{du}{dx} dx$$

$$\int_0^2 x e^{-x} dx = [x \times -e^x]_0^2 - \int_0^2 -e^{-x} \times (1) dx$$

$$\int_0^2 x e^{-x} dx = [-x e^x]_0^2 + \int_0^2 e^{-x} dx$$

$$(-2e^2 - (0)) - (e^2 - (-1)) = 1 - 3e^{-2}$$

Evaluate: $\int_{\pi/4}^{\pi/2} 7x \sin 3x \, dx$

$$u = 7x, \frac{dv}{dx} = \sin 3x, \quad \frac{du}{dx} = 7, \quad v = -\frac{1}{3} \cos 3x$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int u \frac{dv}{dx} dx = \left[(7x) \left(-\frac{1}{3} \cos 3x \right) \right]_{\pi/4}^{\pi/2} - \int -\frac{7}{3} \cos 3x \, dx$$

$$= \left[-\frac{7x}{3} \cos 3x + \frac{7}{9} \sin 3x \right]_{\pi/4}^{\pi/2}$$

$$\text{Top: } -\frac{7}{3} \times \pi/2 \times \cos 3\pi/2 + \frac{7}{9} \sin 3\pi/2$$

$$\cos 3\pi/2 = 0, \sin 3\pi/2 = -1$$

$$\text{Top: } -\frac{7}{9}$$

$$\text{Bottom: } -\frac{7}{3} \times \pi/4 \times \cos 3\pi/4 + \frac{7}{9} \sin 3\pi/4$$

$$\cos 3\pi/4 = -\frac{1}{\sqrt{2}}, \sin 3\pi/4 = \frac{1}{\sqrt{2}}$$

$$\text{Bottom: } \frac{7\pi}{12\sqrt{2}} + \frac{7}{9\sqrt{2}}$$

$$= -\frac{7}{9} - \frac{7\pi}{12\sqrt{2}} - \frac{7}{9\sqrt{2}} \approx -2.62$$

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Integration by substitution

Consider integrating $\int (3x + 1)^4 dx$

We could expand it, but expanding something to the fourth power is a pain.

Instead, we can make a substitution:

Let $y = 3x + 1$, now our integral becomes easier: $\int y^4 dx$

We still have a problem – the dx term!

We can differentiate: $\frac{dy}{dx} = 3$, so $dx = \frac{dy}{3}$

Substitution workings $\int (3x + 1)^4 dx$

$$\int (3x + 1)^4 dx = \int y^4 \frac{1}{3} dy$$

$$\int (3x + 1)^4 dx = \frac{1}{3} \int y^4 dy$$

$$\int (3x + 1)^4 dx = \frac{1}{3} \frac{y^5}{5} + c$$

$$(3x + 1)^4 dx = \frac{y^5}{15} + c = \frac{(3x + 1)^5}{15} + c$$

Second example

Using integration by substitution to find $\int x \sin 2x^2 \, dx$

$$\text{Let: } y = 2x^2, \quad \text{then: } \frac{dy}{dx} = 4x \quad \text{so: } dx = \frac{dy}{4x}$$

$$\int x \sin 2x^2 \, dx = \int x \sin y \frac{1}{4x} \, dy = \int \frac{\sin y}{4} \, dy$$

$$\int \frac{\sin y}{4} \, dy = \frac{\cos y}{4} + c = \frac{\cos(2x^2)}{4} + c$$

Your turn

Try finding $\int t\sqrt{t^2 + 1}dt$ using substitution

$$\text{Let } y = t^2 + 1,$$

$$\text{then } \frac{dy}{dt} = 2t,$$

$$\text{so } dt = \frac{1}{2t} dy$$

$$\int t\sqrt{t^2 + 1}dt = \int t\sqrt{y} \frac{1}{2t} dy = \frac{1}{2} \int y^{0.5} dy$$

$$\frac{1}{2} \int y^{0.5} dy = \frac{1}{2} \times \frac{2}{3} y^{\frac{3}{2}} + c$$

$$\int t\sqrt{t^2 + 1}dt = \frac{1}{3} y^{\frac{3}{2}} + c$$

$$\int t\sqrt{t^2 + 1}dt = \frac{1}{3} (t^2 + 1)^{\frac{3}{2}} + c$$

Substitution with limits

Evaluate: $\int_1^2 \frac{3}{(2t+1)^2} dt$

Let $y = 2t + 1$,

then $\frac{dy}{dt} = 2$,

so $dt = \frac{1}{2} dy$

$$\int \frac{3}{(2t+1)^2} dt = \int \frac{3}{y^2} \times \frac{1}{2} dy$$

Our limits are $t = 1$ and $t = 2$

When $t = 1, y = 3$, and when $t = 2, y = 5$,

$$\text{So } \int_1^2 \frac{3}{(2t+1)^2} dt = \int_3^5 \frac{3}{y^2} \times \frac{1}{2} dy$$

$$= \frac{3}{2} \int_3^5 \frac{1}{y^2} dy = \frac{3}{2} \left[-\frac{1}{y} \right]_3^5$$

$$= \frac{3}{2} \left(-\frac{1}{5} + \frac{1}{3} \right) = 0.2$$

The general solution

$$\int_a^b f(g(x))g'(x)dx, \text{ where } g(x) = u, \int_{g(a)}^{g(b)} f(u)du$$

This is interesting – I've described this as the product of a function and its derivative

We looked at $\int x \sin 2x^2$, where x is half the derivative of x^2

We looked at $\int t\sqrt{t^2 + 1}dt$, where t is half the derivative of t^2

It isn't always obvious that this is the case... $\int_1^2 \frac{3}{(2t+1)^2} dx$

Selecting the right technique

Integration by parts and integration by substitution overlap – you might be able to use either.

We can use Integration by parts to solve integrals where we have two functions multiplied (though we may need to use it multiple times):

$$\int f(x)g(x)$$

(Picking the right one to integrate and the right one to differentiate takes practice)

Generally substitution is easier (fewer ‘moving parts’)

We can use Integration by substitution to solve integrals where one function is multiplied by its derivative:

$$\int f(x)\dot{f}(x)$$

(For definite integrals we need to change the limits according to our substitution)

Chapter 37, p480 in 'Foundation Maths', by Tony Croft